



School of Electronics Engineering (SENSE)

BEVD206L

SEMICONDUCTOR DEVICE MODELLING

COURSE PROJECT

**MODELLING CIGAAFET AND OBTAINING DC
PARAMETERS WITH SELF-HEATING EFFECTS**

Submitted To

Dr. Lakshmi B

Submitted By

24BVD1006-Sonakshi Agrawal

24BVD1033 – Sanjana Chejeti

24BVD1069 – Katakam Shrihita

24BVD1076 – Vyshnavi.E

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ABSTRACT

This report presents a comprehensive Sentaurus TCAD simulation study of 8 nm nano-wire Gate-All-Around Field-Effect Transistors (NW GAAFETs) in Core-Insulator GAAFET (CIGAAFET) configuration. The key direct current (DC) performance parameters — including threshold voltage (V_{th}), transconductance (g_m), Drain-Induced Barrier Lowering (DIBL), Subthreshold Swing (SS), Ion/Ioff ratio, and output conductance (g_o) — are extracted. The CIGAAFET, with a SiO₂ core-insulator swept from 1 nm to 7 nm radius, systematically improves the Ion/Ioff ratio up to 1.68×10^9 at RCI = 7 nm while maintaining acceptable on-current levels. It retains its performance advantage over the standard GAAFET even when SHEs are included. These findings confirm the CIGAAFET as a compelling architecture for low-power, energy-efficient nanoscale electronics. This report also explores key difference between traditional junctionless FINFETs and GAAFETS thus highlighting their limitations and future scope.

Keywords: *GAAFET, CIGAAFET, Sentaurus TCAD, Self-Heating Effects, Short Channel Effects, Subthreshold Swing, DIBL.*

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INTRODUCTION

The relentless miniaturisation of silicon transistors, guided by Moore's Law, has been the cornerstone of the semiconductor industry for over five decades. As transistor dimensions are scaled to the sub-10 nm regime, traditional planar MOSFET architectures encounter severe performance degradations arising from Short Channel Effects (SCEs). The principal among these is Drain-Induced Barrier Lowering (DIBL), where the depletion region extending from the drain begins to modulate the potential barrier at the source-channel junction, effectively reducing the gate's control over the channel. Additional SCEs including velocity saturation, subthreshold slope degradation, and increased off-state leakage current further erode device performance at these scales.

The transition from planar MOSFETs to three-dimensional FinFET architectures offered a significant reprieve by wrapping the gate around three sides of a raised silicon fin, enabling multiple inversion layers and improved electrostatic control. However, as device dimensions approach and fall below 7 nm, even FinFETs suffer from resurgent SCEs, prompting the semiconductor industry to explore the logical extension of the multi-gate concept: the Gate-All-Around Field-Effect Transistor (GAAFET).

In the GAAFET, the gate electrode completely surrounds the channel on all sides, providing maximum electrostatic control and effectively suppressing SCEs. The two primary channel geometries under investigation are the nano-wire (NW), which has a cylindrical cross-section, and the nano-sheet (NS), which has a rectangular cross-section. While GAAFETs offer superior gate control, they introduce a critical new challenge: Self-Heating Effects (SHEs). The silicon channel, completely encased in a thermally insulating high-k dielectric (typically HfO_2), cannot efficiently dissipate the Joule heat generated during device operation. This thermal confinement, compounded by the dramatically reduced thermal conductivity of silicon at the nanoscale, leads to elevated channel temperatures that degrade carrier mobility and reduce drive current.

This report presents a detailed simulation study conducted using the Synopsys Sentaurus TCAD suite (Version W-2024.09) on CIGAAFET configuration based on an 8 nm radius cylindrical nano-wire:

- Core-Insulator GAAFET (CIGAAFET): A novel variant introducing a solid SiO_2 cylinder at the centre of the silicon channel, converting it into a tubular structure to suppress off-state leakage current.

For the configuration, standard DC transfer and output characteristics are simulated and key performance parameters are extracted. The CIGAAFET's core-insulator radius (RCI) is taken as 5nm and -current and off-current suppression is evaluated.

DEVICE STRUCTURE AND SIMULATION SETUP

2.1 CIGAAFET Structure

The Core-Insulator Gate-All-Around Field-Effect Transistor (CIGAAFET) considered in this work is based on a cylindrical nanowire architecture in which a coaxial silicon dioxide (SiO_2) core is embedded within the silicon channel, forming a hollow channel structure. The device is constructed using Sentaurus Structure Editor (SDE) with the transport direction defined along the z-axis.

- The **silicon channel has a radius of 8 nm** and extends over a **length of 200 nm**, while a central SiO_2 **core of radius 5 nm** is introduced along the same axis, resulting in an effective silicon shell thickness of **approximately 3 nm**. This core insulator suppresses the low-electric-field bulk conduction path through the center of the channel, thereby reducing off-state leakage current while preserving the high-field inversion layer at the semiconductor–oxide interface.
- The source and drain regions are modelled as cylindrical silicon extensions on either side of the channel, each with a radius of 8 nm. The **source spans from -10 nm to 0 nm**, and the **drain extends from 200 nm to 210 nm** along the channel axis. Both regions are heavily doped with **arsenic at a concentration of $1 \times 10^{19} \text{ cm}^{-3}$** to form n^+ contacts, while the channel is lightly doped with **boron at $1 \times 10^{16} \text{ cm}^{-3}$** , resulting in an $n^+ \text{--} p \text{--} n^+$ device configuration.
- Surrounding the channel is a **high-k gate dielectric layer of HfO_2 with an outer radius of 10 nm**, providing strong electrostatic control over the channel. Electrical contacts are defined for the gate, source, and drain, with the gate assigned a **work function of 4.55 eV**. Mesh refinement is applied at critical interfaces, including the source–channel, channel–drain, and channel–oxide regions, to accurately capture variations in electric field and carrier distribution within the nanoscale device.

2.2 Simulation Sweeps

The transfer characteristics of the device in the linear region are evaluated.

- The drain voltage (VD) is fixed at a low bias of 0.01 V, and the gate voltage (VG) is swept from 0 V to 2 V in incremental steps to analyze the variation of drain current with gate bias.

To ensure numerical stability and convergence, the gate voltage sweep is performed stepwise using the DoZero command, which reinitializes the solution at each bias point and prevents convergence issues arising from nonlinear transport and temperature-dependent effects. The simulation setup, combined with fine mesh refinement at key interfaces, enables accurate evaluation of the electrical characteristics of the proposed device

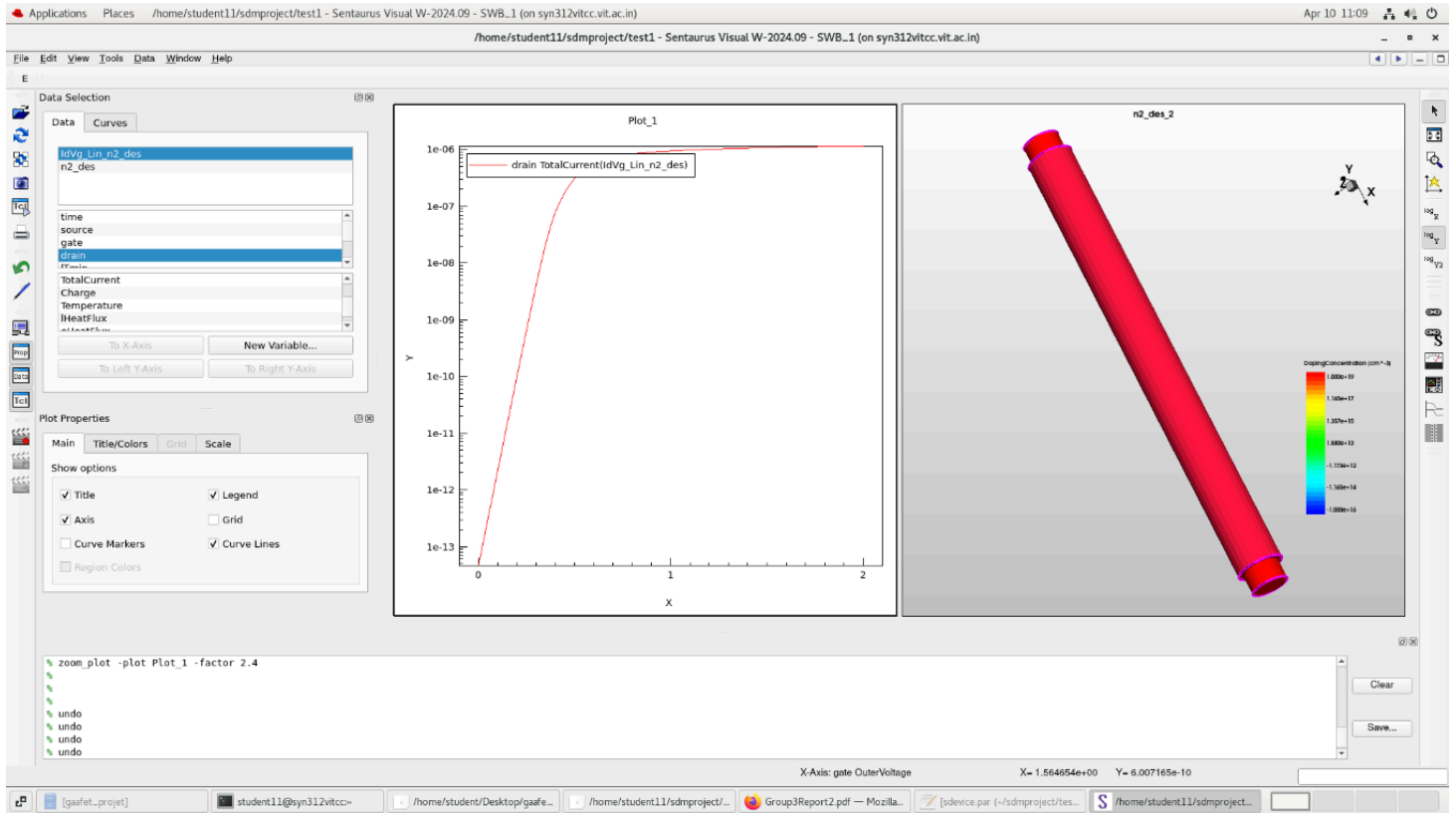


Fig1: CIGFET structure with Id-Vg graph (logy)

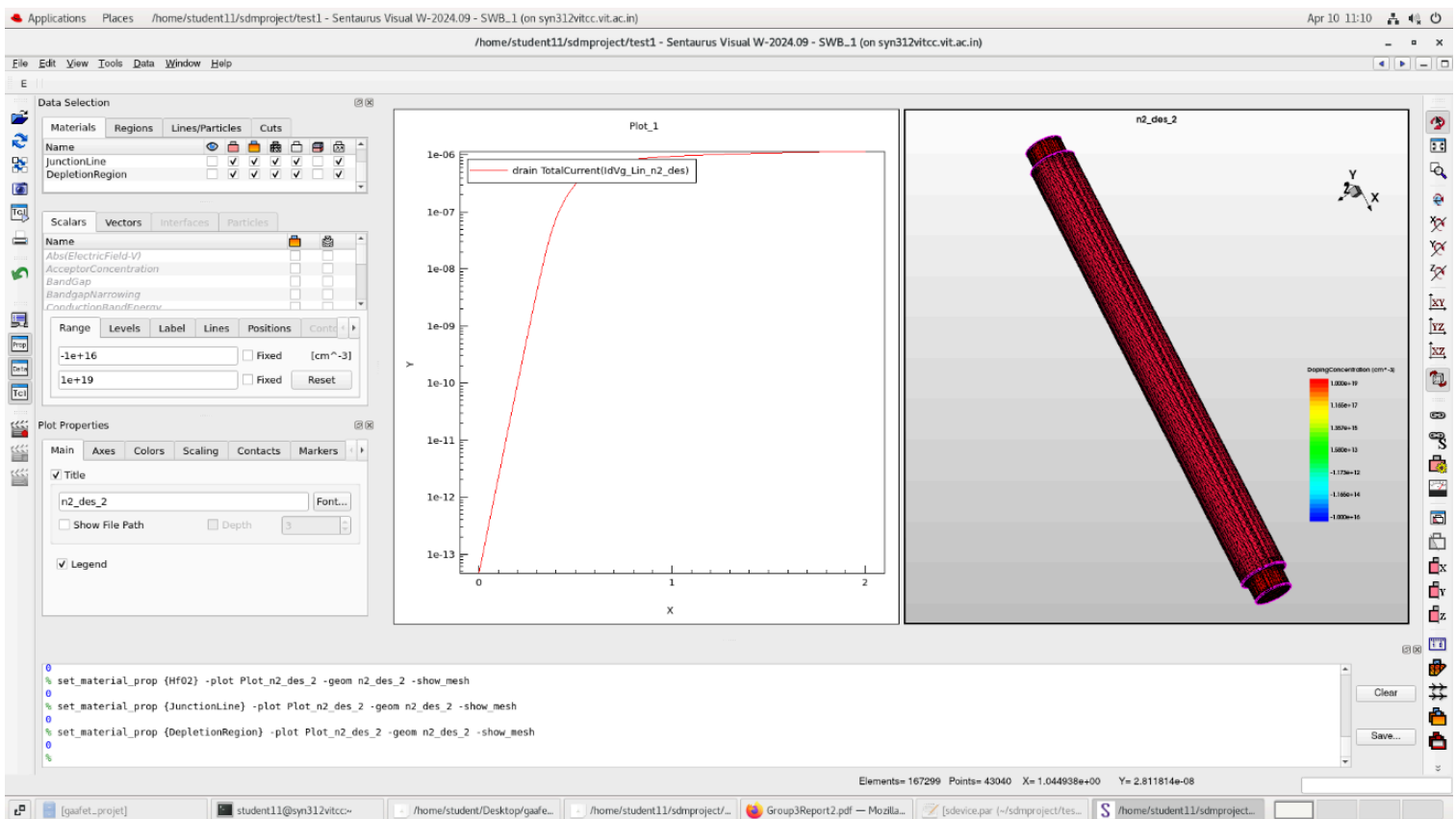


Fig2: CIGFET structure with meshing, along with Id-Vg graph (logy)

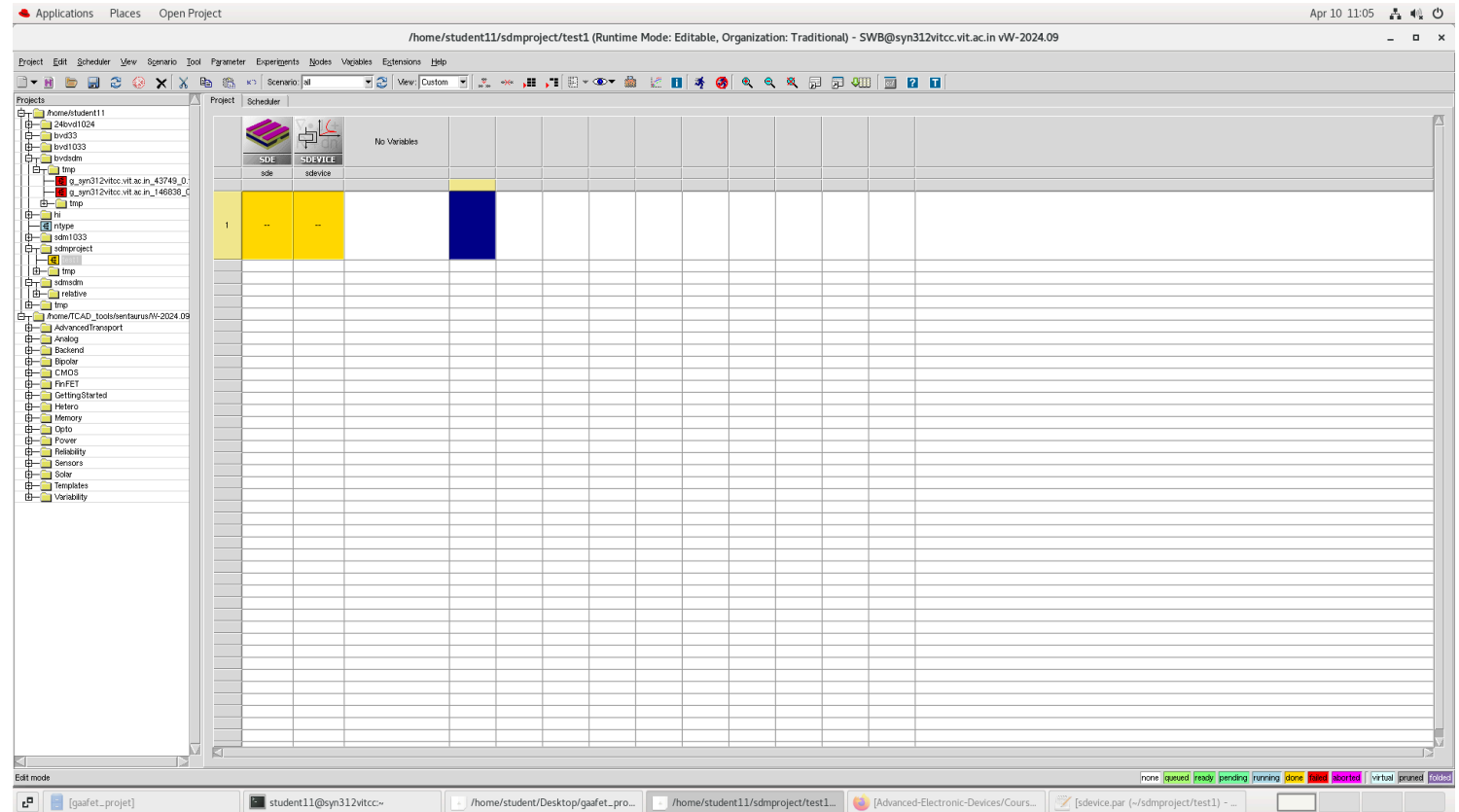


Fig3: swb workbench with SDE and SDEVICE nodes.

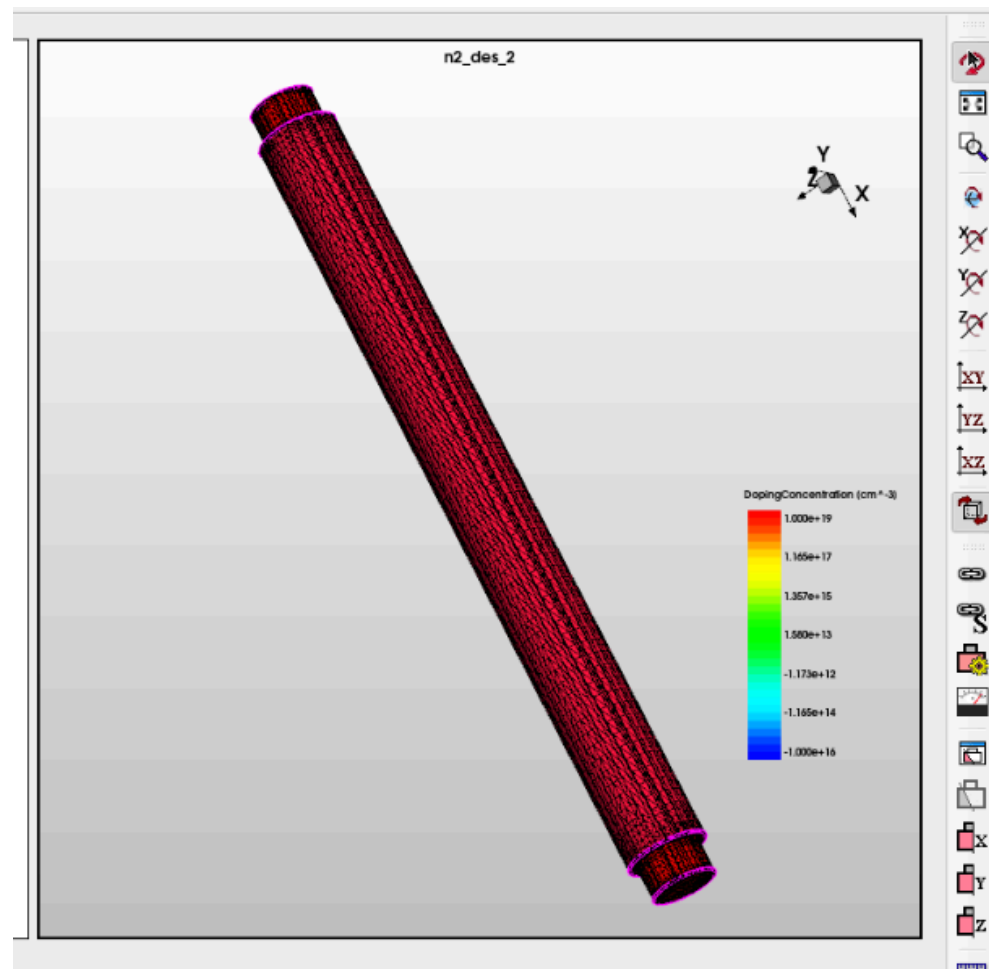


Fig4: Id-Vg graph for CIGAFET along with meshing structure

PARAMETER EXTRACTION METHODOLOGY

All key DC parameters are extracted from the simulated I-V characteristics using the Inspect post-processing tool within the Sentaurus TCAD framework, and verified using MATLAB polynomial fitting. The extraction procedures are described below.

3.1 Threshold Voltage — Linear Region ($V_{th,lin}$)

The linear-region threshold voltage is extracted using the transconductance extrapolation (G_m -max) method. The drain current I_{DS} is differentiated with respect to V_{GS} to obtain $g_m = dI_{DS}/dV_{GS}$. The voltage at which g_m reaches its maximum ($V_{GS,max}$) is identified. The tangent line to the I_{DS} – V_{GS} curve at $V_{GS,max}$ is then extrapolated to intersect the V_{GS} axis ($I_{DS} = 0$), giving $V_{th,lin}$. In the Inspect tool, this is implemented using the

$$VT1 \text{ macro: } curve_1 = VT1(TotalCurrent_drain),$$

which returned $V_{th,lin} = 0.4114$ V (raw tool value), refined to 0.39 V after polynomial fitting.

3.2 Threshold Voltage — Saturation Region ($V_{th,sat}$)

The saturation-region threshold voltage is extracted using the square-root method. The square root of I_{DS} (i.e., $\sqrt{I_{DS}}$) is plotted against V_{GS} . The maximum slope point on this curve is identified and the tangent extrapolated to the V_{GS} axis.

3.3 DIBL

Drain-Induced Barrier Lowering is calculated using:

$$DIBL = |V_{th,lin} - V_{th,sat}| / |V_{DS,lin} - V_{DS,sat}| \text{ (mV/V)},$$

where $V_{DS,lin} = 0.01$ V and $V_{DS,sat} = 1.5$ V.

3.4 Subthreshold Swing (SS)

Subthreshold swing is computed as:

$$SS = (d \log_{10}(I_{DS}) / dV_{GS})^{-1} \text{ (mV/dec)}.$$

In Inspect, this is realised by creating $curve_2 = \log_{10}(TotalCurrent_drain)$ and differentiating to find the slope in the subthreshold region.

3.5 Ion and Ioff

The on-voltage and off-voltage are defined as:

$$V_{DS,ON} = V_{DS,sat} + V_{th,sat} - |V_{th,sat}| \text{ and } V_{DS,OFF} = V_{th,sat} - |V_{th,sat}|.$$

I_{on} and I_{off} are read directly from the transfer characteristic at $V_{GS} = V_{DS,ON}$ and $V_{GS} = V_{DS,OFF}$ respectively.

3.6 Output Conductance (g_o)

The maximum output conductance is determined from the I_D - V_{DS} characteristic as: $g_o = \max(dI_{DS}/dV_{DS})$ in the range $V_{DS} = 1.5$ V to 2 V. The fourth derived curve ($curve_4 = 1/(curve_2)$) of the differentiated log-current is used in conjunction with the tangent macro to extract this value.

RESULTS AND DISCUSSION

4.1 Effect of Core-Insulator Radius on DC Parameters

The CIGAAFET introduces a SiO₂ cylinder of radius RCI at the centre of the silicon channel, leaving a silicon shell of thickness (8 - RCI) nm as the active channel region. Table 2 summarises the complete set of extracted DC parameters for RCI value of 5 nm. **Work function = 4.55**

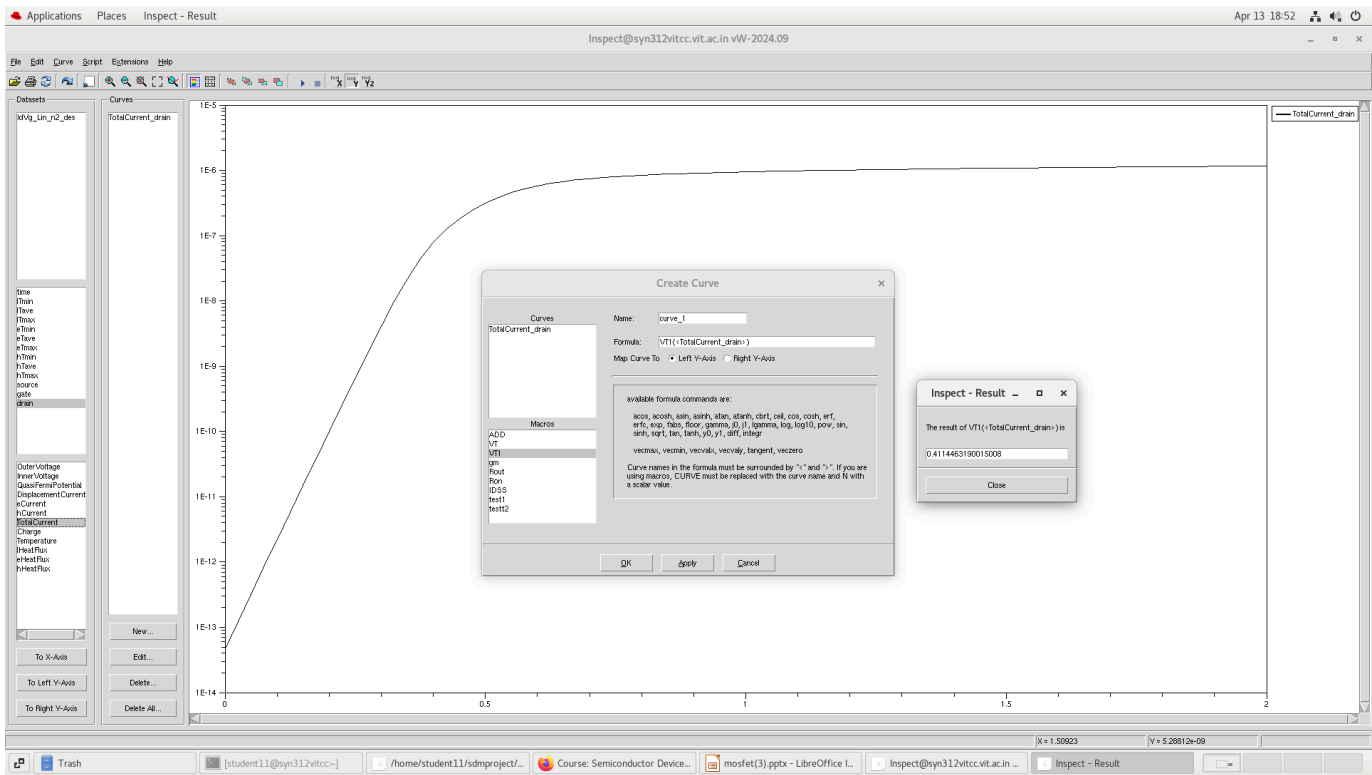
Table 2: DC Parameters for GAAFET and CIGAAFET (RCI = 5nm) Without SHEs (Taken from references)

Parameter	GAAFET	CI: 5nm
V _{th_lin} (V)	0.39	0.40
V _{th_sat} (V)	0.36	0.37
DIBL (mV/V)	20.61	16.34
SS (mV/Dec)	59.61	59.59
I _{on} (A)	6.36E-05	6.10E-05
I _{off} (A)	1.37E-13	8.28E-14
I _{on} /I _{off}	4.64E+08	7.37E+08
g ₀ (μS)	0.84	0.45

Table 3: DC Parameters for GAAFET and CIGAAFET (RCI = 1–7 nm) With SHEs (Simulated)

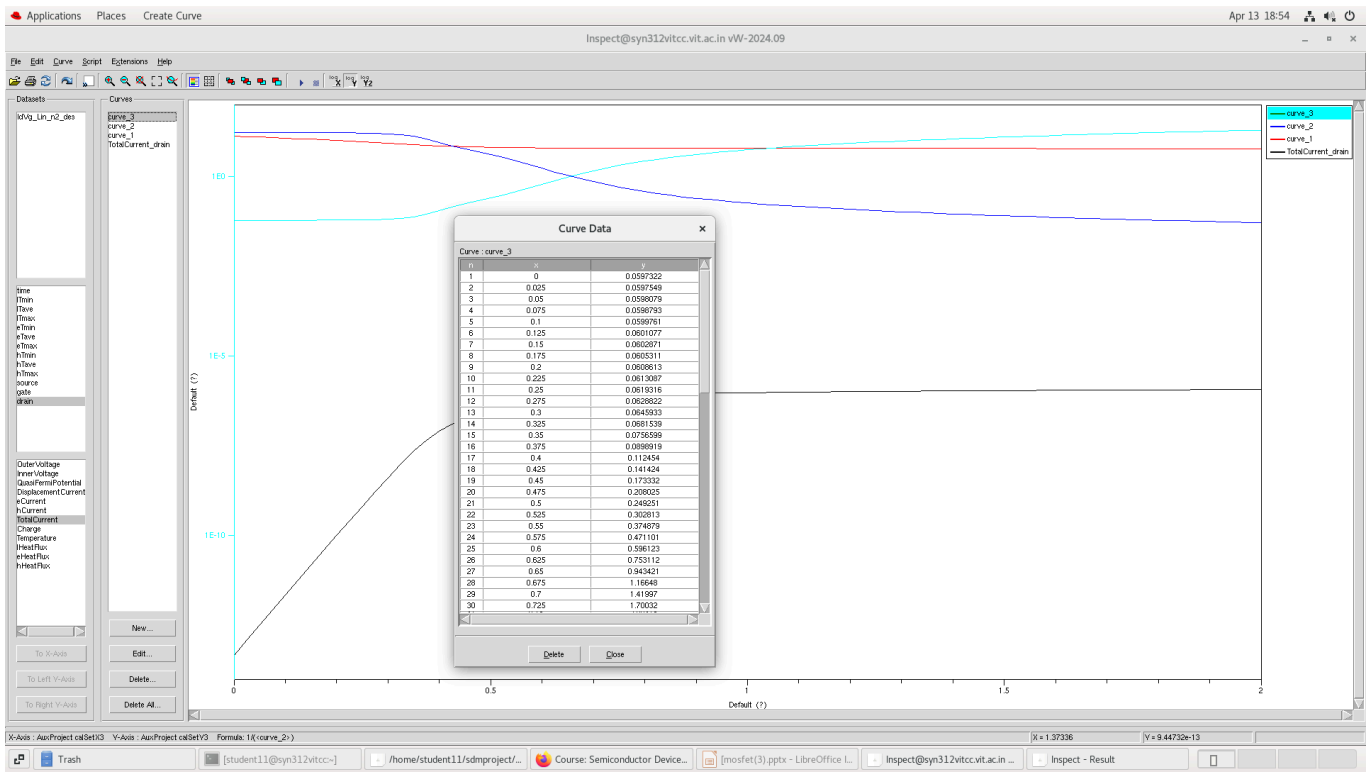
Parameter	GAAFET	CI: 5nm
V _{th_lin} (V)	0.39	0.41
V _{th_sat} (V)	0.36	0.37
DIBL (mV/V)	19.32	17.68
SS (mV/Dec)	59.65	59.67
I _{on} (A)	5.66E-05	5.33E-05
I _{off} (A)	1.34E-13	8.05E-14
I _{on} /I _{off}	4.21E+08	6.63E+08
g ₀ (μS)	0.89	0.63

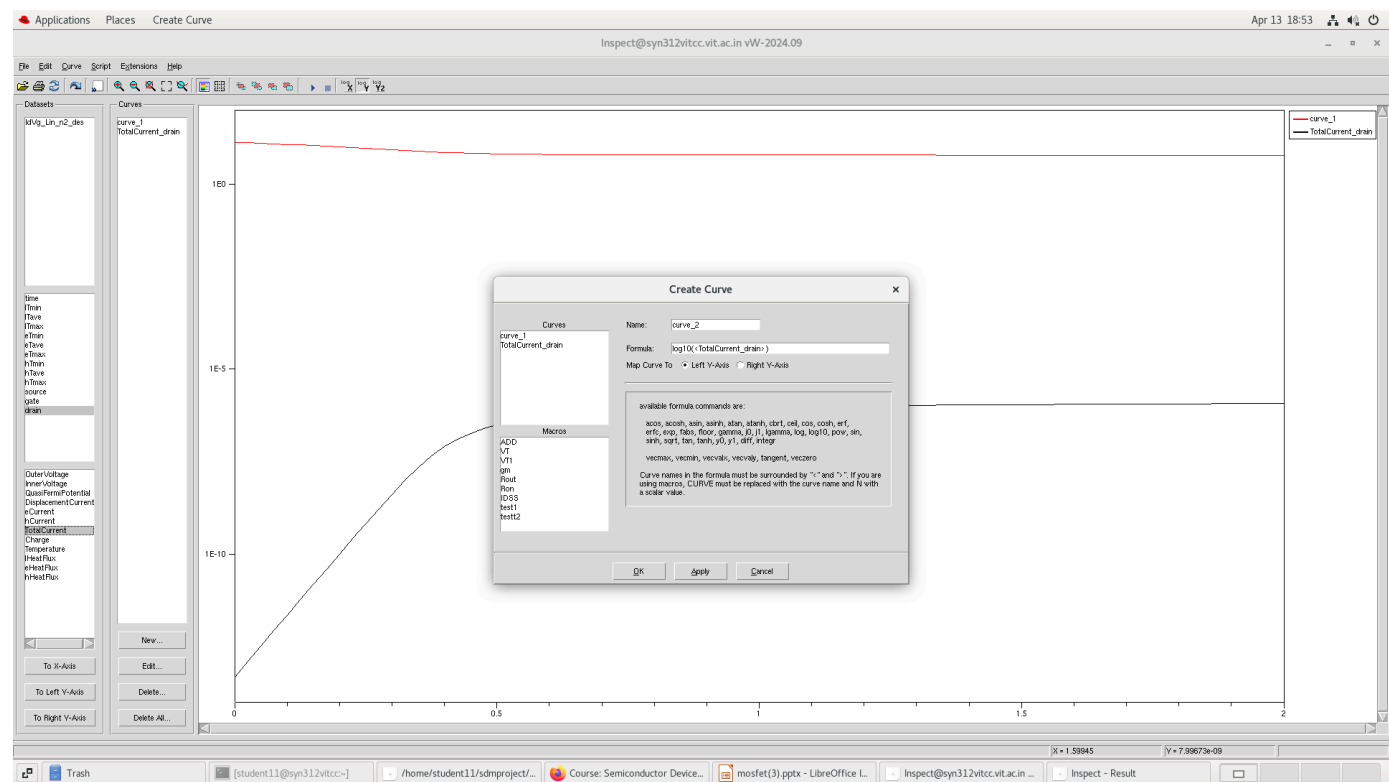
4.2 Threshold Voltage Trends



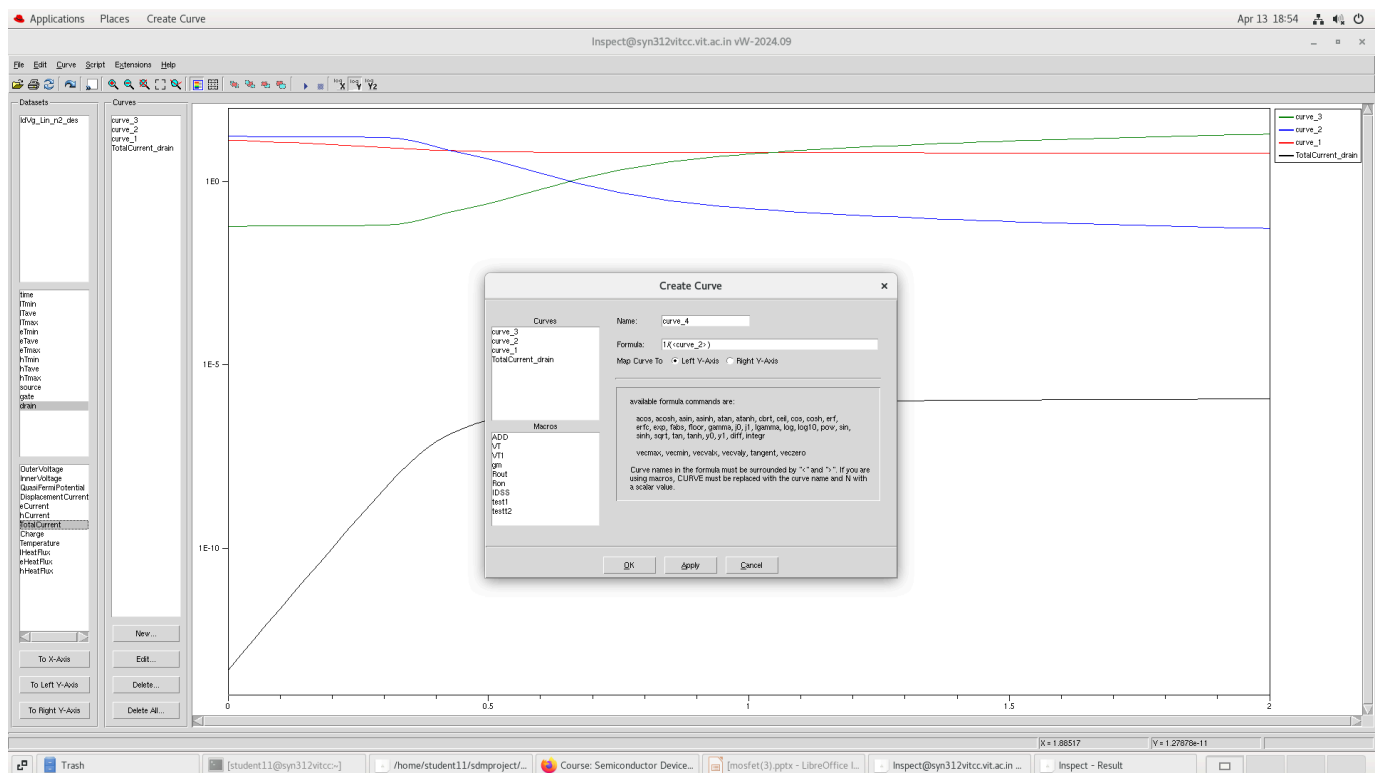
4.3 DIBL Improvement and Subthreshold Swing =0.06

Work function =4.5





4.4 Output Conductance



DISCUSSION

5.1 Comparison of Device Architectures

Feature	Junctionless FinFET	CIGAAFET (Cylindrical Gate-All-Around FET)
Structure	Fin-shaped channel with gate on three sides	Cylindrical nanowire channel fully surrounded by gate
Gate Control	Good electrostatic control (3D)	Excellent control (360° gate coverage)
Channel Doping	Uniformly heavily doped (no junctions)	Typically lightly doped or intrinsic
Fabrication Complexity	Easier (no abrupt junctions required)	More complex (nanowire fabrication)
Short Channel Effects (SCE)	Reduced compared to planar MOSFET	Strongly suppressed due to full gate control
Leakage Current	Low	Very low
Carrier Transport	Bulk conduction	Surface + volume conduction (better control)
Scalability	Good up to sub-10 nm	Excellent for sub-5 nm nodes
Threshold Voltage Control	Controlled via work function and doping	Controlled via gate material and geometry
Performance	Moderate improvement over FinFET	Higher performance and better switching
Applications	Low-power, simpler ICs	Advanced nodes, high-performance VLSI

Key Insights

- **Junctionless FinFETs** simplify fabrication and reduce variability by eliminating junctions.
- **CIGAAFETs** provide **superior electrostatic control**, making them more suitable for **future nanoscale technologies**.

5.1.1 Junctionless FinFET

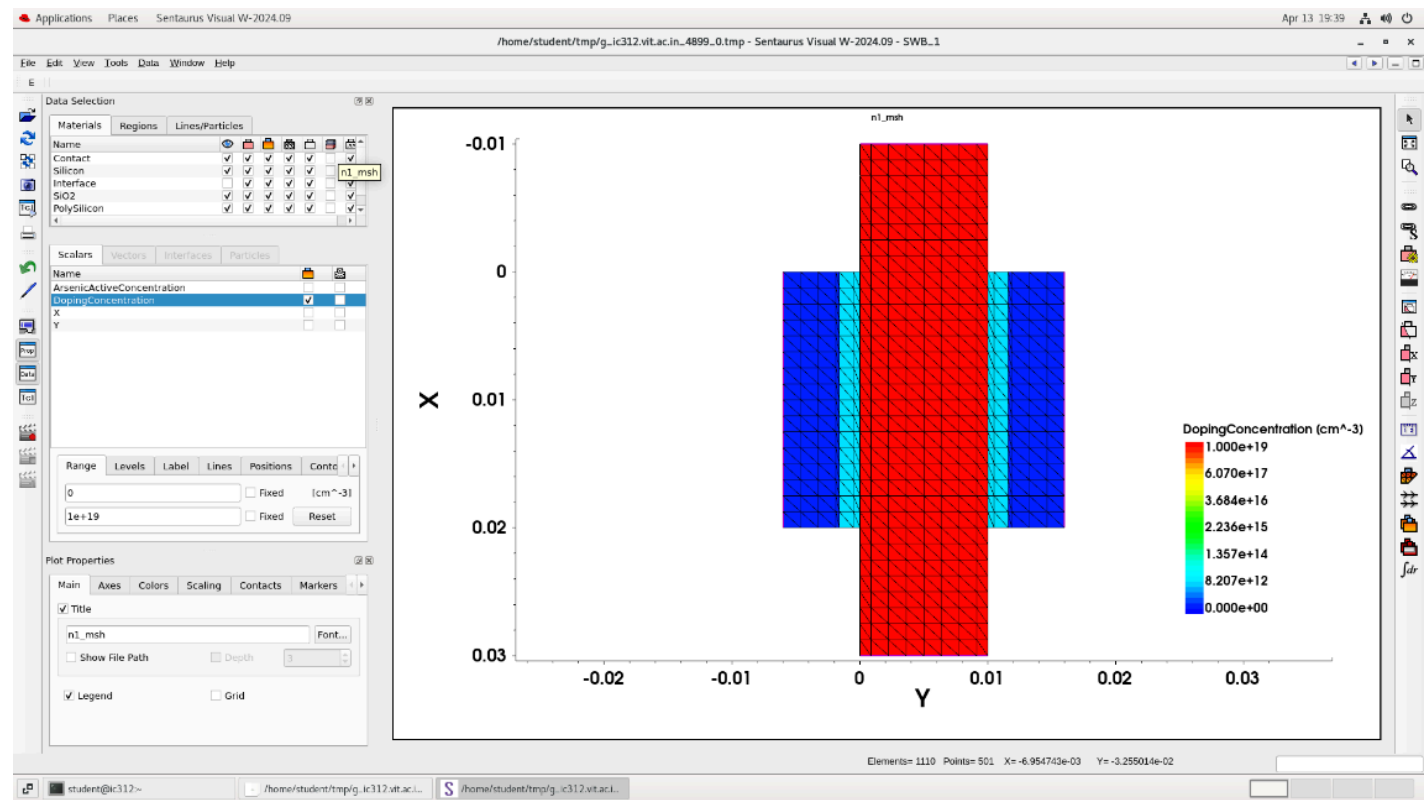


Fig4: Junctionless FINFET structure along with meshing

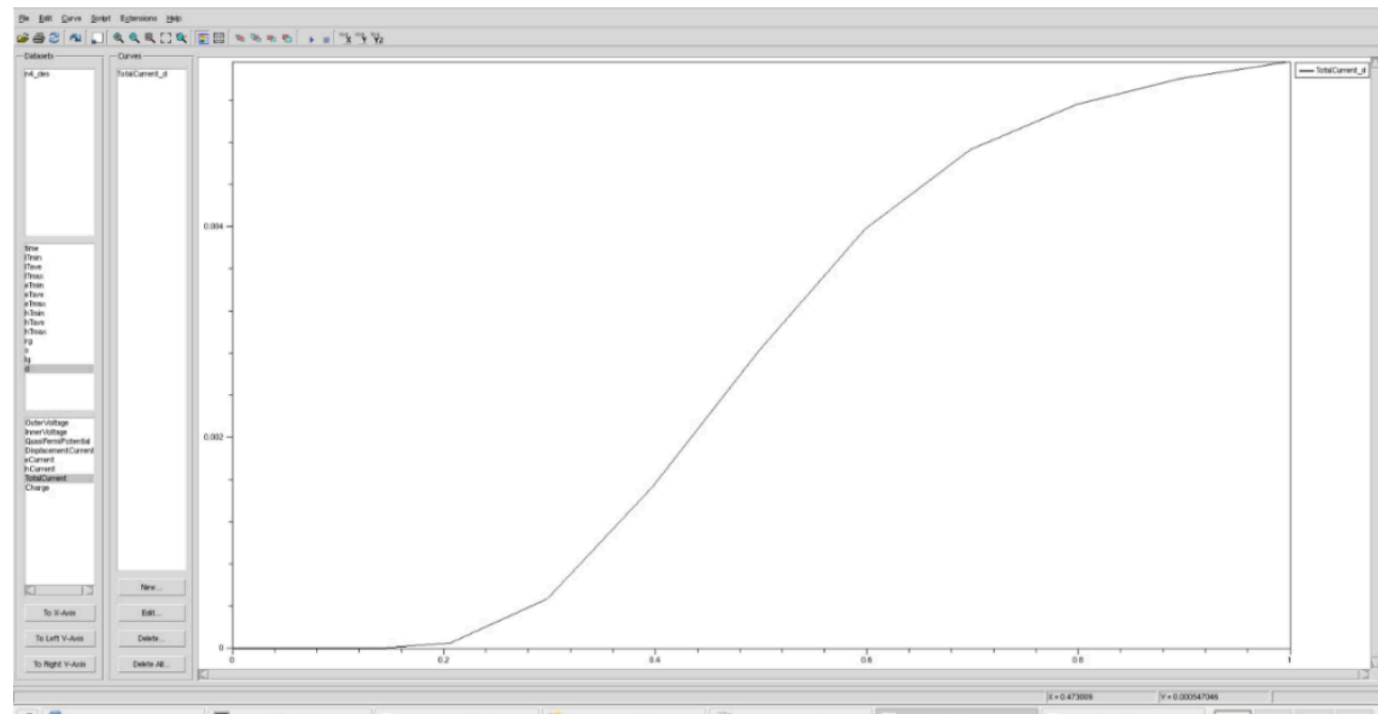


Fig5: I_d - V_g graph for junctionless FINFET

5.2 Physical Insights

The cylindrical nano-wire geometry provides a key physical insight: the electron density in the on-state (as shown in Figure 12 of the reference paper, reproduced in the TCAD visualisation) is maximum near the gate oxide interface and minimum at the channel centre. This radial concentration of the inversion layer means that removing the central volume of the channel (by introducing the SiO₂ core) has minimal impact on the on-current while dramatically reducing the off-state bulk current path. This elegant geometric solution is what makes the CIGAAFET concept so compelling.

The thermal analysis reveals that the cylindrical GAAFET geometry faces a fundamental thermal bottleneck: the HfO₂ gate dielectric ($\kappa \approx 0.014 \text{ W cm}^{-1} \text{ K}^{-1}$) acts as a thermal insulator that confines Joule heat to the silicon channel. At the nanoscale, silicon's own thermal conductivity is reduced by approximately two orders of magnitude relative to bulk due to phonon boundary scattering, further exacerbating the problem. The resulting 82 K temperature rise during saturation operation represents a significant self-heating penalty that must be considered in circuit-level simulations.

5.3 Limitations and Future Work

Several important physical effects have not been included in the present simulations and represent areas for future work:

- Surface roughness scattering: The gate-oxide interface roughness in GAAFETs degrades carrier mobility near the surface, which is precisely where the inversion layer forms. Including the Coulomb scattering and surface roughness components of the Phillips Unified Mobility model would likely reduce I_{on} by an additional 5–15%.
- Phonon boundary scattering: At 8 nm dimensions, phonon mean free paths are comparable to device dimensions, making quantum corrections to thermal transport important.
- Quantum confinement: In a 1 nm silicon shell ($RCI = 7 \text{ nm}$), strong quantum confinement modifies the subband structure and effective density of states, potentially shifting threshold voltage by tens of millivolts.
- Correct device dimensions: The simulated device has an 8 nm radius (16 nm diameter) rather than the originally specified 4 nm radius (8 nm diameter). Devices at the correct 4 nm radius would exhibit stronger quantum confinement, better gate control, and potentially different SHE profiles.

CONCLUSION

These findings corroborate the GAAFET's status as a highly effective solution for maintaining transistor performance at the 8 nm node and below, and support the CIGAAFET as a promising evolution of the GAAFET concept for energy-efficient applications. The widespread industry adoption of GAAFET-based designs (e.g., Samsung's MBCFET, Intel's RibbonFET) underscores the practical relevance of the phenomena studied in this work.

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APPENDIX

SDE CODE

```
(set! process-up-direction "+z")
(sdegeo:set-auto-region-naming OFF)
(sdegeo:set-default-boolean "ABA")
(sdegeo:create-cylinder (position 0 0 0) (position 0.2 0 0) 0.005 "SiO2" "CoreIns")
(sdegeo:set-default-boolean "BAB")
(sdegeo:create-cylinder (position 0 0 0) (position 0.2 0 0) 0.008 "Silicon" "Channel")
(sdegeo:set-default-boolean "ABA")
(sdegeo:create-cylinder (position -0.01 0 0) (position 0 0 0) 0.008 "Silicon" "Source")
(sdegeo:create-cylinder (position 0.2 0 0) (position 0.21 0 0) 0.008 "Silicon" "Drain")
(sdegeo:set-default-boolean "BAB")
(sdegeo:create-cylinder (position -0.001 0 0) (position 0.201 0 0) 0.01 "HfO2" "Oxide")
(sdegeo:define-contact-set "gate" 4 (color:rgb 1 0 0) "##")
(sdegeo:define-contact-set "source" 4 (color:rgb 0 1 0) "||")
(sdegeo:define-contact-set "drain" 4 (color:rgb 0 0 1) "==")
(sdegeo:set-current-contact-set "gate")
(sdegeo:define-3d-contact (list (car (find-face-id (position 0.1 0.0 -0.01)))) "gate")
(sdegeo:set-current-contact-set "source")
(sdegeo:define-3d-contact (list (car (find-face-id (position -0.01 0 0)))) "source")
(sdegeo:set-current-contact-set "drain")
(sdegeo:define-3d-contact (list (car (find-face-id (position 0.21 0 0)))) "drain")
(sdedr:define-constant-profile "Const.Channel" "BoronActiveConcentration" 1e16)
(sdedr:define-constant-profile-region "PlaceCD.Channel" "Const.Channel" "Channel")
(sdedr:define-constant-profile "Const.Source" "ArsenicActiveConcentration" 1e+19)
(sdedr:define-constant-profile-region "PlaceCD.Source" "Const.Source" "Source")
(sdedr:define-constant-profile "Const.Drain" "ArsenicActiveConcentration" 1e+19)
(sdedr:define-constant-profile-region "PlaceCD.Drain" "Const.Drain" "Drain")
#(sdedr:define-refeval-window "BaseLine.Channel_left" "Rectangle" (position -0.012 -0.008 -0.008) (position -0.012
0.008 0.008) )
#(sdedr:define-analytical-profile-placement "PlaceAP.Channel_left" "Gauss.Channel_left" "BaseLine.Channel_left"
"Both" "NoReplace" "Eval")
#(sdedr:define-gaussian-profile "Gauss.Channel_left" "ArsenicActiveConcentration" "PeakPos" 0 "PeakVal" 1e+19
"ValueAtDepth" 1e+17 "Depth" 0.010 "Gauss" "Factor" 0.8)
#(sdedr:define-refeval-window "BaseLine.Channel_right" "Rectangle" (position 0.212 -0.008 -0.008) (position 0.212
0.008 0.008) )
```

```

#(sdedr:define-analytical-profile-placement "PlaceAP.Channel_right" "Gauss.Channel_right"
"BaseLine.Channel_right" "Both" "NoReplace" "Eval")
#(sdedr:define-gaussian-profile "Gauss.Channel_right" "ArsenicActiveConcentration" "PeakPos" 0 "PeakVal" 1e+19
"ValueAtDepth" 1e+17 "Depth" 0.01 "Gauss" "Factor" 0.8)
>(sdedr:define-refinement-size "RefinementDefinition_1" 0.002 0.002 0.002 0.002 0.002 0.002 )
>(sdedr:define-refinement-placement "RefinementPlacement_1" "RefinementDefinition_1" (list "region" "Source" ) )
#t
>(sdedr:define-refinement-function "RefinementDefinition_1" "MaxLenInt" "Source" "Channel" 0.001 1.5
"UseRegionNames")
#t
>(sdedr:define-refinement-function "RefinementDefinition_1" "MaxLenInt" "Source" "source" 0.001 1.5
"UseRegionNames")
#t
>
>(sdedr:define-refinement-size "RefinementDefinition_2" 0.005 0.002 0.002 0.005 0.002 0.002 )
#t
>(sdedr:define-refinement-placement "RefinementPlacement_2" "RefinementDefinition_2" (list "region" "Channel" ) )
#t
>(sdedr:define-refinement-function "RefinementDefinition_2" "MaxLenInt" "Channel" "Source" 0.001 1.5
"UseRegionNames")
#t
>(sdedr:define-refinement-function "RefinementDefinition_2" "MaxLenInt" "Channel" "Drain" 0.001 1.5
"UseRegionNames")
#t
>(sdedr:define-refinement-function "RefinementDefinition_2" "MaxLenInt" "Channel" "Oxide" 0.001 1.5
"UseRegionNames")
#t
>(sdedr:define-refinement-size "RefinementDefinition_3" 0.002 0.002 0.002 0.002 0.002 0.002 )
#t
>(sdedr:define-refinement-placement "RefinementPlacement_3" "RefinementDefinition_3" (list "region" "Drain" ) )
#t
>(sdedr:define-refinement-function "RefinementDefinition_3" "MaxLenInt" "Drain" "Channel" 0.001 1.5
"UseRegionNames")
#t
>(sdedr:define-refinement-function "RefinementDefinition_3" "MaxLenInt" "Drain" "drain" 0.001 1.5
"UseRegionNames")
#t
# (sde:set-meshing-command "snmesh -a -c boxmethod")

```

```
# (sdedr::append-cmd-file "")
# (sde:build-mesh "snmesh" "-a -c boxmethod" "n@node@")
```

SDEVICE CODE

```
File {
CMIPath = ". Libcmi"
Grid= "n@node|sde@_msh.tdr"
Output= "@log@"
Plot= "n@node@_des.tdr"
Current="n@node@_des.plt"
Parameter= "@parameter@"
}

Electrode {
{ Name="source" Voltage= 0 }
{ Name="drain" Voltage= 0 }
{ Name="gate" WorkFunction= 4.55 Voltage= 0 }
}

*For Schottky S & D contact cfr Ali video
*Electrode {
*{ Name="source" Material= "Metal" Voltage= 0 }
*{ Name="drain" Material= "Metal" Voltage= 0 }
*{ Name="gate" WorkFunction= 4.55 Voltage= 0 }
*}

Thermode {
{ Name="source" Temperature=300 SurfaceResistance=1e-9 }
{ Name="drain" Temperature=300 SurfaceResistance=1e-9 }
}

Physics {
Temperature= 300
Hydrodynamic(eTemperature)
}

Physics(Material= "Silicon") {
eThermionic
hThermionic
Mobility (
PhuMob
Enormal
```

```

eHighFieldsat(CarrierTempDrive)
hHighFieldsat(GradQuasiFermi) )
*eAvalanche(CarrierTempDrive) hAvalanche(GradQuasiFermi)
EffectiveIntrinsicDensity(BandGapNarrowing (OldSlotboom))
Recombination( SRH(DopingDependence))
ThermalConductivity( Constant Conductivity )
}
*For PeltierHeat activation in Ohmic contact
Physics(Electrode="source"){ MSPeltierHeat}
Physics(Electrode="drain"){ MSPeltierHeat}
*For Schottky contact
*Physics(MaterialInterface= "Metal/Silicon"){
*Schottky
*MSPeltierHeat
*Recombination(eBarrierTunneling hBarrierTunneling)
*BarrierLowering(Full)
*}
Plot {
Doping DonorConcentration AcceptorConcentration
BandGap BandGapNarrowing ElectronAffinity
ConductionBandEnergy ValenceBandEnergy
eQuasiFermiEnergy hQuasiFermiEnergy
eDensity hDensity
EffectiveIntrinsicDensity IntrinsicDensity
ElectricField/Vector eEparallel hEparallel
eQuasiFermi hQuasiFermi
egradQuasiFermi hgradQuasiFermi
Potential Doping SpaceCharge
SRH
eCurrent hCurrent
eMobility hMobility eVelocity hVelocity
Temperature
eTemperature hTemperature
ThermalConductivity
*Nonlocal
*BarrierTunneling
}
*For Schottky

```

```

*Math(MaterialInterface= "Metal/Silicon"){
*NonLocal(Length= 1e-7)
*Digits(NonLocal=3)
*EnergyResolution(NonLocal)= 0.001
Math {
Number_of_Threads= 20
Iterations= 20000
Extrapolate
*RelErrControl
RhsAndUpdateConvergence
AcceptNewtonParameter ( -RhsAndUpdateConvergence RhsMin = 1.5e-1 UpdateScale = 1e-2)
EquilibriumSolution(Iterations=1000 Digits=7 LineSearchDamping=0.001)
}
Solve {
Coupled{Poisson }
Coupled{Poisson Electron Hole }
Coupled(Iterations= 150) {Poisson Electron Hole
eTemperature
Temperature
}

Quasistationary(
AcceptNewtonParameter (ReferenceStep = 0.1)
DoZero
MaxStep= 0.05
Decrement = 2
Increment= 1e10
MinStep= 1e-9
Goal { Name="drain" Voltage= 0.01 }
)
{ Coupled(Iterations= 50) { Poisson Electron Hole
eTemperature
Temperature
} }

Quasistationary(
AcceptNewtonParameter (ReferenceStep = 0.1)
MaxStep= 0.05
Decrement = 2

```

```

Increment= 1e10
MinStep= 1e-9
Goal { Name="gate" Voltage= 0 }
)
{ Coupled(Iterations= 50) { Poisson Electron Hole
eTemperature
Temperature
} }
NewCurrentFile="IdVg_Lin_"
Quasistationary(
AcceptNewtonParameter (ReferenceStep = 0.1)
DoZero
MaxStep= 0.05
Decrement = 2
Increment= 1e10
MinStep= 1e-9
Goal { Name="gate" Voltage= 0.5 }
)
{ Coupled(Iterations= 50) { Poisson Electron Hole
eTemperature
Temperature
} }
Quasistationary(
AcceptNewtonParameter (ReferenceStep = 0.1)
DoZero
MaxStep= 0.05
Decrement = 2
Increment= 1e10
MinStep= 1e-9
Goal { Name="gate" Voltage= 1 }
)
{ Coupled(Iterations= 50) { Poisson Electron Hole
eTemperature
Temperature
} }
Quasistationary(
AcceptNewtonParameter (ReferenceStep = 0.1)
DoZero

```

```

MaxStep= 0.05
Decrement = 2
Increment= 1e10
MinStep= 1e-9
Goal { Name="gate" Voltage= 1.5 }
)
{ Coupled(Iterations= 50) { Poisson Electron Hole
eTemperature
Temperature
} }
Quasistationary(
AcceptNewtonParameter (ReferenceStep = 0.1)
DoZero
MaxStep= 0.05
Decrement = 2
Increment= 1e10
MinStep= 1e-9
Goal { Name="gate" Voltage= 2 }
)
{ Coupled(Iterations= 50) { Poisson Electron Hole
eTemperature
Temperature
} }

```

